

SEBASTIAN CHIRINOS NEGRON

AI / Data Engineer | Python · Vector Databases · Embeddings

Arequipa, Peru (UTC-5) | Available for Remote Work - US & EU time zones | English C1

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sebastian-cn-portfolio.vercel.app

PROFESSIONAL SUMMARY

Software Engineer moving into AI and data engineering, with 2+ years of production Python and PostgreSQL behind five shipped products. Builds semantic retrieval systems on vector databases (ChromaDB, Sentence-Transformers), evaluates embedding models against real corpora using cosine similarity, PCA and t-SNE, and ships on-device computer-vision inference with quantized TensorFlow Lite models. Combines that with the ability to put a model behind a working product - API, database and interface. Works daily across the Python analysis stack: Pandas, NumPy and Jupyter. Fluent English (C1), full daily overlap with US and European time zones.

PROFESSIONAL EXPERIENCE

Software Engineer (Contract / Freelance)

Jan 2024 - Present

Independent - Remote | Clients: Calitop Services, Lo Exacto, Revolt Laptop, GEOTOP

- Built and evaluated **semantic retrieval systems on vector databases** (ChromaDB + Sentence-Transformers): corpus design, embedding generation, persistence, similarity search and metadata filtering.
- Designed and queried **PostgreSQL** schemas backing production systems, including a point-of-sale platform handling **100+ transactions per day** across a **500+ SKU** inventory.
- Built a Python (Flask) document service storing and indexing **700+ technical certificates** in S3-compatible object storage for GEOTOP, deployed on a self-provisioned Google Compute Engine VM behind Nginx, with QR-based retrieval from physical equipment.
- Trained and **quantized a TensorFlow Lite computer-vision model** for fully offline anemia screening on Android, running inference on-device with no network dependency.

DATA & MACHINE LEARNING PROJECTS

Semantic Corpus Analysis - Embedding Model Evaluation - github.com/BastleyNait/LAB02-TABD

Analyzed a 36-document Spanish technical corpus spanning three domains (epidemiological surveillance, vector databases, AI). Produced a 36x36 cosine-similarity heatmap plus PCA and t-SNE projections to contrast global variance with local cluster structure, and implemented semantic outlier detection at a 0.6 threshold. Benchmarked all-MiniLM-L6-v2 against paraphrase-multilingual-MiniLM-L12-v2 and recommended the multilingual model on measured gains in Spanish technical-synonym detection.

Python, ChromaDB, Sentence-Transformers, scikit-learn, PCA, t-SNE, Matplotlib, Jupyter

Semantic Course Recommender - Vector Database - github.com/BastleyNait/LAB01-04-TABD

Built a recommendation engine over a curated course corpus using ChromaDB persistent storage and Sentence-Transformers embeddings, supporting interest-based recommendation, similarity search and skill-profile matching with metadata filtering. Documented measured limitations - generic embeddings missing local domain terminology, strict equality filtering - and specified hybrid BM25 + vector retrieval and a relevance-feedback loop as the next iteration.

Python, ChromaDB, Sentence-Transformers, semantic search, RAG patterns

Anemivision - Offline On-Device Anemia Screening - github.com/BastleyNait/ANEMIVISION

Native Android application running anemia detection entirely on-device with a quantized computer-vision model, built for field use with no network connectivity.

Kotlin, Android, TensorFlow Lite, PyTorch, model quantization

TECHNICAL SKILLS

Data & ML: Python, Pandas, NumPy, Jupyter, Sentence-Transformers, ChromaDB, PyTorch, TensorFlow Lite, PCA, t-SNE, Embeddings, Semantic Search, RAG, Model Evaluation

Databases: PostgreSQL, SQL, Vector Databases (ChromaDB), MongoDB, Redis, Schema Design, Query Optimization

Backend: Python, FastAPI, Flask, Django, Node.js, SQLAlchemy, REST APIs, GraphQL, JWT Authentication, RBAC, Next.js API Routes, Microservices

Cloud & DevOps: Google Cloud Platform (Compute Engine), Vercel, Nginx, Linux (Ubuntu), Docker, Docker Compose, CI/CD, Git, GitHub, GitHub Actions, S3-compatible Object Storage, Reverse Proxy

EDUCATION & LANGUAGES

Data Engineering on Google Cloud - Professional Certificate (in progress)

Systems Engineering - Universidad Nacional de San Agustín de Arequipa (UNSA), Peru

Languages: Spanish (Native) · English - C1 Advanced, professional working proficiency